

# National University of Computer and Emerging Sciences, Lahore Campus



|             |                  |              |           |
|-------------|------------------|--------------|-----------|
| Course:     | Embedded Systems | Course Code: | EE419     |
| Program:    | BS (EE)          | Semester:    | Fall 2018 |
| Duration:   | 1 Hour           | Total Marks: | 60        |
| Paper Date: | 16-Nov-18        | Weight:      | 15 %      |
| Section:    | ALL              | Page(s):     | 4         |
| Exam:       | Midterm - II     |              |           |
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## Instruction/Notes:

- The exam is OPEN book and OPEN notes.
- Use only black or blue pen to solve the paper.
- Write your answers in the space provided. You may use extra sheet for rough work only. Do not attach it.
- If need arises, make valid assumptions and clearly mention it with your answer

## Question # 1

CLO – 2.5, 3.2, 3.5 [20 marks]

- 5 A. What is the maximum delay that can be added using systick timer if clock source is 12MHz?  
Show complete calculations. [5 Marks]

$$\frac{1}{f} = t = \frac{1}{12M} = 83.33ns$$

$$2^{24} = 16777216$$

$$1 \text{ cycle time} = 83.33ns$$

24 bit register for

$$\text{delay} = 83.33ns \times 2^{24} = 1.398s$$

- 5 B. How many bits are used in UART baudrate generator to set complete baud rate? If system clock is 20MHz, required baud rate is 115200, what should be loaded in generator if default clock divider value is used. Show complete calculations. [5 Marks]

22 bits are used to set UART.

$$\text{Baudrate} = 115200$$

$$f_{osc} = 20MHz$$

$$\text{divider} = \frac{20MHz}{16 \times 115200} = 10.850$$

$$UART1BRD = 10$$

$$UARTFBRD = (0.850) \times 2^6 = 54$$

- 4 C. We want to control intensity of led (connected with PF3) driven by 1KHz wave generator 3 of pwm module1. Assume all initializations are done except LOAD register. Set these registers for 20% duty cycle of 1KHz wave. Now write code for this duty cycle by 20% after every 20msec. void Delay\_20ms() is available to you. cycle reaches 100%, reset it to 20%. TM4C is running on 16MHz clock. [10 Marks]



What is the maximum delay that can be added using systick timer if clock source is 12MHz?  
Show complete calculations. [5 Marks]

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$$\text{Baudrate} = 115200$$

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$$\text{UARTIBRD} = 10$$

$$\text{UARTFBRD} = (0.850) \times 2^6 = 54$$

4 C. We want to control intensity of led (connected with PF3) driven by 1KHz wave, using generator 3 of pwm module1. Assume all initializations are done except LOAD and CMP register. Set these registers for 20% duty cycle of 1KHz wave. Now write code for increasing this duty cycle by 20% after every 20msec. void Delay\_20ms() is available to you. After duty cycle reaches 100%, reset it to 20%. TM4C is running on 16MHz clock. [10 Marks]

$$\text{Load} = \frac{1}{16M} \times \frac{1}{m} = 16k$$

$$\text{CMPA} = \frac{100 - 20}{100} \times 16k = 128000$$



PWM1 - 3 - Load - R = 16000;

## Question # 2

CLO - 3.3-3.4 [40 marks]

We need to design a system to count number of people entering a hall. There are two doors of the hall, each connected with a sensor. Door 1 is connected with sensor1 which gives a low pulse on detecting a person. Door2 is connected with sensor 2 which is different than sensor 1 and outputs a high pulse on detecting a person. These pulses are of very low width. This embedded system is also connected with a display via SSI. The total number of persons in the hall are displayed whenever a push button is pressed (it gives '0' when pressed). We are using TM4C controller and using following pins:

- PF0 with sensor 1 - Door 1 (low pulse)
- PF1 with sensor 2 - Door 2 (high pulse)
- PA2 with Push button
- SSI2 module

Use interrupts for each input. PA2 should have highest priority. There is no need of SSI2 interrupt. The display connected to the system receives one byte of data at a time at the rate of 250K. Use minium possible value for SSI clock prescale. The display can print a character at a time. TM4C is working on 16MHz.

Assume the capacity of the hall is 500. People are not allowed to enter when hall is full.

Do the following tasks:

### A. Interfacing sensors: [10 Marks]

switches

i. Sensors are interfaced using external interrupts. Write subroutine void sensorInit for configuring interrupts for sensors PF0, PF1

void sensorInit() {

sys

PIO\_PORTF\_IS\_R & = ~0x03;

IO\_PORTF\_IBE\_R & =

4



- PF1 with sensor 2 — Door 1 (low pulse)
- PA2 with Push button — Door 2 (high pulse)
- SSI2 module

Use interrupts for each input. PA2 should have highest priority. There is no need of SSI2 interrupt. The display connected to the system receives one byte of data at a time at the rate of 250K. Use minimum possible value for SSI clock prescale. The display can print a character at a time. TM4C is working on 16MHz.

Assume the capacity of the hall is 500. People are not allowed to enter when hall is full.

Do the following tasks:

A. Interfacing sensors: [10 Marks]

- i. Sensors are interfaced using external interrupts. Write subroutine void sensorInit() for configuring interrupts for sensors PFO, PF1 switches

void sensorInit()

~~sys~~

```

GPIO_PORTF_IS_R &= ~0x03;
GPIO_PORTF_IBE_R &=
GPIO_PORTF_IEN_R &= ~0x03;
GPIO_PORTF_ICR_R = 0x03;
GPIO_PORTF_IM_R |= 0x03; // interrupt on PFO and PF1
NVIC_ENO_R = 0x40000000;
Enable_interrupts();

```



ii. Write handler routine(s) for part i

```
static unsigned char enter = 0;
volatile int readback;
while (GPIO_PORTF-MIS-R & 0x01) {
    if (GPIO_PORTF-MIS-R & 0x01) {
        count = count++;
        GPIO_PORTA-ICR = 0x01;
    }
    else {
        (GPIO_PORTF-MIS-R & 0x02)
        { count = count + 1; }
        GPIO_PORTA-ICR = 0x02; }
}
```

B. Interface Push button: [15 Marks]

i. Configure interrupt for push button

void PushButtonInit() {

```
GPIO_PORTA-IS-R = 0x04;
GPIO_PORTA-IBE-R = 0x04;
GPIO_PORTA-IEV-R = 0x04;
GPIO_PORTA-ICR-R = 0x04;
GPIO_PORTA-IM-R = 0x04;
```

```
NVIC-ENO-R = 0x000000001;
} Enable Interrupts;
```

ii. Write handler routine for this interrupt void PushButtonHandler()

```
int i;
static unsigned char count = 0;
volatile int readback;
while (GPIO_PORTA-ISR-R & 0x01) {
    if (GPIO_PORTA-ISR-R & 0x01) {
        count = count++;
        GPIO_PORTA-ICR-R = 0x01;
    }
    else {
        (GPIO_PORTA-ISR-R & 0x02)
        { count = count + 1; }
        GPIO_PORTA-ICR-R = 0x02; }
}
```



```
GPIO_PORTA_ICR_R = 0x04;  
GPIO_PORTA_IM_R |= 0x04;
```

```
NVIC_ENO_R = 0x00000001;  
EnableInterrupts();
```

```
}
```

ii. Write handler routine for this interrupt `void PushButtonHandler()`

```
int i;  
static unsigned char count = 0;  
volatile int readback;  
while (GPIO_PORTA_MIS_R != 0)  
if (GPIO_PORTA_MIS_R & 0x04)  
{  
    if (count < 500)  
    {  
        count  
        SS12Write(count);  
    }  
    else  
    {  
        SS12Write('f');  
        // f for full → 500  
    }  
    GPIO_PORTA_ICR_R = 0x04  
    readback = GPIO_PORTA_ICR_R;  
}
```



C. Displaying total count of persons: [10 Marks]

- i. Write subroutine void SSI2Init() to configure SSIx module. Before that show all the calculations for setting data rate

SYSCTL - RCGCSSI - R1 = 4;

SSI2 - CR1 - R = 0

SSI2 - CR0 - R = 0x00002007;

PB4, 5, 6, 7

CLK, FSS, RX, TX

~~Data rate = 250k~~

~~fosc = 16MHz~~

~~Divider =  $\frac{16\text{MHz}}{16 \times 250 \times 10^3}$~~

~~Divider = 4~~

Prescaler = 2

fSSI = (16MHz) / (2 \* (1 + SCR))

SCR =  $\left(\frac{16\text{M}}{2 \times 250\text{k}}\right)^{-1} = 31$

fSSI =

SCR = 0x20 0x02  
250k data rate.

ii.

Write subroutine void SSI2Write(unsigned char data) to send 8 bit data over pin

PB7



ii.

Write subroutine void SSI2Write(unsigned char data) to send 8 bit data over Tx pin

PB7

```
void SSI2Write(unsigned char data)
{
    GPIO_PORTB_DATA_R |= 0x20;
    SSI2_DR = data;
    while (SSI2_SR & 0x10);
    GPIO_PORTB_DATA_R |= 0x20;
}
```

D. Write main ( ) function which should call all required routines [5 Marks]

```
void main(void)
{
    unsigned char a;
    void sensorInit();
    void PushbuttonInit();
    void PushbuttonHandler();
    void SSI2init();
    SSI2write(a);
}
```

```
while(1)
{
}
```

negative marks for this



Q#02.

PushButtonHandler();

// assume karna tha k 3 digit value  
aari hai (500 given)  
iska mod lna tha ADC wala  
kaam;

aur phr ek ek digit display karana  
tha SS12write() key since yh syf  
ek character kara sakta ~~sto~~ send.

$$\text{data rate} = \frac{16M}{2 \times (1 + SCR)}$$

↑  
minimum value = 2

$$SCR = 31$$

$$CRO-R = 0x\textcircled{1F}\underline{07}$$



Mid #02

Q#01 (c)

100 - duty cycle = off time

~~16M~~ ~~16000~~ ~~16000~~ ~~16000~~

$$\frac{16M}{1k} = 16000 = \text{load}$$

CMP mein off cycle load karayi hai.

$$\text{initial value in CMP} = \frac{16000 \times 20}{100}$$

→ (book k hisaab sey)

$$\text{CMP} = 0.8 \times 16000 = 12800$$

$$\text{let } x = 12800$$

After every 20 ms, DC ko 20% inc karna. → (Duty cycle)

while(1) {

$$\text{cmp} = x$$

$$x = x - 3200;$$

if ( $x \leq 0$ )

$$x = 12800;$$

delay - 20ms();

}

For on time ;

add 3200

$$x = x + 3200$$

calculations:

$$\begin{aligned} 20\% \text{ DC} &= 12800 \\ 40\% \text{ DC} &= 9600 \\ 60\% \text{ DC} &= 6400 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{diff of } 3200$$

3200

→ 40% Duty cycle:

$$0.6 \times 16000 = 9600$$

→ 60% duty cycle:

$$0.4 \times 16000 = 6400$$